Production strict-past conditional Hessian and weighted-collar boundary

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

first issued 2026-08-03 · this version issued 2026-08-03 · v1.0

1. Purpose and scope: identify the actual conditional owner

R-151 proves a strict local Hessian gap at zero deterministic strict past for the complete linear $p : 2p$ endpoint. R-152 derives an affine-past chain rule, a conditional absolute collar, and the abstract nonlinear and multi-root acceptance gates. It deliberately stops before identifying the conditional operator on actual production incidences. The present result performs that identification for the last fresh root and records the strongest consequence that follows without inventing a new owner.

There are four conclusions.

1. After conditioning on the strict past, the fresh current covariance cancels exactly against its part of the A7 trace counterterm. The remaining endpoint contains the past covariance with a minus sign. Its first variation and complete symmetric bilinear Hessian are displayed below.
2. The terminal sixth-power and Cameron–Martin source Hessians have the exact factors 9/10 and 9/10. A curved control chart adds the first variation applied to the chart acceleration; fixed-law differentiation does not create an unlabelled Malliavin term.
3. Replacing the R-152 spatial essential suprema by spatial L^2 norms gives a strictly more flexible pathwise collar. The exact A1 source symbol also gives the control-generated mean bound
$$\|\nabla m_h\|_2^2 < \frac{41}{20}\|h_-\|_{\mathcal{S}}^2.$$
Neither statement controls the revealed Gaussian past unconditionally.
4. A nondegenerate derivative-active Gaussian past has unbounded support, so neither the R-152 absolute collar nor the new L^2 collar can hold with a fixed almost-sure threshold in general. Event-localized predictable controls show that an all-directions weighted-average version is equivalent to the fiberwise bound. The signed covariance-normal completion remains necessary.

This is a scoped T4 boundary. It does not construct the earlier-root feedback, complete forest/low/balanced assembly, or the multi-root production matrix. T-050, A13, Nelson, and Sector A remain open. No phase or PDE verdict follows.

2. Exact strict-past reduction

Fix the A7 six-real convention, positive coefficient floor, finite cutoff, and one common real-even regulator. Condition on the strict-past sigma-field $\mathcal{F}_-$. For almost every past realization write

$$W = m + X, \qquad V_i = n_i + Y_i,$$

where (X, Y_i) is the centered fresh Gaussian field-current pair. Assume the production same-point identities

$$\mathbb{E}_f[XY_i^T] = 0, \qquad \mathbb{E}_f[Y_iY_i^T] = \Gamma_{>,i}, \qquad \Gamma_{\Lambda,i} = \Gamma_{<,i} + \Gamma_{>,i}. \tag{2.1}$$

For a whitened source direction a, let the actual final-root incidences be

$$z_a = S_x a, \qquad u_{i,a} = S_{v,i}a. \tag{2.2}$$

Production links these quantities:

$$n_i = \partial_i m, \qquad S_{v,i} = \partial_i S_x. \tag{2.3}$$

They are not freely variable model parameters.

Let $B = C_6^* C_6$ be the complete positive Gram coefficient and define

$$\overline{B}_a(x) = \mathbb{E}_f B(m + X + z_a). \tag{2.4}$$

Theorem 2.1 (future-current/trace cancellation)

The conditional A7 endpoint for the declared final fresh root is exactly

$$\boxed{\mathcal{R}_-(a) = \frac{1}{2}\int \sum_i \left[(n_i + u_{i,a})^T \overline{B}_a (n_i + u_{i,a}) - \mathrm{Tr}(\overline{B}_a \Gamma_{<,i})\right] dx.} \tag{2.5}$$

Proof. Before integrating the fresh current, the ith A7 contribution is

$$\frac{1}{2}\mathbb{E}_f \left[(n_i + u_i + Y_i)^T B(m + X + z)(n_i + u_i + Y_i) - \mathrm{Tr}\{B(m + X + z)\Gamma_{\Lambda,i}\}\right]. \tag{2.6}$$

Joint Gaussianity and the first identity in (2.1) make X and Y_i independent. Thus the linear Y_i terms vanish and

$$\mathbb{E}_f[Y_i^T B(m + X + z)Y_i] = \mathrm{Tr}(\overline{B}_z \Gamma_{>,i}). \tag{2.7}$$

The future trace cancels against the future part of $\Gamma_{\Lambda,i}$, leaving

$$\mathrm{Tr}(\overline{B}_z \Gamma_{>,i}) - \mathrm{Tr}\{\overline{B}_z(\Gamma_{<,i} + \Gamma_{>,i})\} = -\,\mathrm{Tr}(\overline{B}_z \Gamma_{<,i}). \tag{2.8}$$

This proves (2.5). □

The cancellation is signed and owner-complete for this conditional endpoint. It must occur before any absolute-value estimate. The result would change if the regulator were asymmetric, the same-point field-current covariance were nonzero, or the conditional law moved with a.

3. First variation, Hessian, and form criterion

At $a = 0$ put $W = m + X$. For directions α, β, abbreviate

$$z_\alpha = S_x\alpha, \quad u_{i,\alpha} = S_{v,i}\alpha, \quad \overline{DB[z_\alpha]} = \mathbb{E}_f DB(W)[z_\alpha], \tag{3.1}$$

and similarly for D^2B.

Proposition 3.1 (complete first and second variations)

The first variation is

$$\boxed{\begin{aligned} D\mathcal{R}_-(0)[\alpha] = \int \sum_i \Big[& u_{i,\alpha}^T \overline{B}_0 n_i \\ & + \frac{1}{2}\operatorname{Tr}\Big(\overline{DB[z_\alpha]}(n_i n_i^T - \Gamma_{<,i})\Big)\Big] dx. \end{aligned}} \tag{3.2}$$

The symmetric bilinear Hessian is

$$\boxed{\begin{aligned} D^2\mathcal{R}_-(0)[\alpha,\beta] = \int \sum_i \Big[& u_{i,\alpha}^T \overline{B}_0 u_{i,\beta} \\ & + u_{i,\alpha}^T \overline{DB[z_\beta]} n_i + u_{i,\beta}^T \overline{DB[z_\alpha]} n_i \\ & + \frac{1}{2}\operatorname{Tr}\Big(\overline{D^2B[z_\alpha, z_\beta]}(n_i n_i^T - \Gamma_{<,i})\Big)\Big] dx. \end{aligned}} \tag{3.3}$$

On the diagonal, the two cross orientations combine to coefficient two. The past-covariance contribution retains coefficient $-1/2$.

Proof. Differentiate the one scalar endpoint (2.5). The quadratic current factor contributes two first-current orientations and one current-current term. The field dependence contributes one DB and one D^2B. Polarization gives (3.3), including the two separately oriented cross terms. □

Equation (3.2) is generally nonzero. Positive Hessian curvature alone does not prove that the base point is stationary or a local minimum.

Exact sixth-power and source owners

For

$$\mathcal{S}_6(a) = \frac{3}{20}\int \mathbb{E}_f |W + z_a|^6 dx,$$

the exact derivatives are

$$D\mathcal{S}_6(0)[\alpha] = \frac{9}{10}\int \mathbb{E}_f[|W|^4 W \cdot z_\alpha] dx, \tag{3.4}$$

and

$$\boxed{\begin{aligned} D^2\mathcal{S}_6(0)[\alpha,\beta] = \frac{9}{10}\int \mathbb{E}_f\big[& |W|^4 z_\alpha \cdot z_\beta \\ & + 4|W|^2 (W\cdot z_\alpha)(W \cdot z_\beta)\big] dx. \end{aligned}} \tag{3.5}$$

This Hessian is positive semidefinite, but it need not be strictly positive. In whitened source coordinates the Cameron–Martin action is

$$\mathcal{C}(a) = \frac{9}{20}|a|^2, \qquad D^2\mathcal{C} = \frac{9}{10}\mathrm{I}. \tag{3.6}$$

For nonwhitened coordinates, I is replaced by the actual source Gram.

Let K_E and K_6 denote the conditional endpoint and sixth-power forms without the source term. The requested one-tenth curvature gap is exactly

$$\boxed{K_E + K_6 \succeq -\frac{4}{5}\mathrm{I} \quad \Longleftrightarrow \quad K_E + K_6 + \frac{9}{10}\mathrm{I} \succeq \frac{1}{10}\mathrm{I}.} \tag{3.7}$$

This is stronger than mere nonnegative curvature, whose endpoint-plus-sixth threshold is $-9\mathrm{I}/10$.

Connection and domain firewall

For a curved source path

$$a(t) = a_0 + t\alpha + \frac{1}{2}t^2\eta + o(t^2),$$

the chain rule is

$$\frac{d^2}{dt^2}\mathcal{A}(a(t))\big|_{t=0} = D^2\mathcal{A}(a_0)[\alpha,\alpha] + D\mathcal{A}(a_0)[\eta]. \tag{3.8}$$

The source connection is $9a_0 \cdot \eta/10$. Endpoint and sixth-power connections need not vanish. Gaussian-source pullback has its own term $DF[LD_\xi^2 h]$; no $D_\xi\phi$ term belongs to the fixed-law Hessian (3.3).

For the strict-past measurable matrix $K = K_E + K_6$, the natural absolute form domain is

$$\mathcal{D}_{|K|^{1/2}} = \{\phi : \mathbb{E}|\phi|^2 + \mathbb{E}[\phi^T|K|\phi] < \infty\}. \tag{3.9}$$

At finite cutoff and fixed floor, bounded localized simple controls form the declared core. If it contains $q\mathbf{1}_E$ for rational q and $E \in \mathcal{F}_-$, then

$$\mathbb{E}[\phi^T(K + 4\mathrm{I}/5)\phi] \geq 0 \quad \text{for every core } \phi \tag{3.10}$$

is equivalent to $K \succeq -4\mathrm{I}/5$ almost surely. If a fiberwise rational direction fails, localize it to the positive-measure failure event. One may not declare the unrestricted form closed and semibounded before this lower bound is proved.

4. Spatially weighted collar

Conditioned on the past, define the unnormalized spatial quantities

$$\begin{aligned} A_n^2 &= \int \sum_i |n_i|^2 dx, \\ A_W^2 &= \int \mathbb{E}_f|m + X|^2 dx, \\ A_Y^2 &= \int \mathbb{E}_f \sum_i |Y_i|^2 dx. \end{aligned} \tag{4.1}$$

Let V be the torus volume, $\lambda_2 = \lambda_{\min} A(2p)$, and retain the exact R-130 envelopes

$$L_6 = \frac{1143}{250P}, \qquad H_6 = \frac{7083}{500P}, \qquad P = 4 + 10^{-12}. \tag{4.2}$$

Theorem 4.1 (conditional spatial L2 collar)

Define

$$\boxed{\begin{aligned} C_{2,p} = {} & \frac{8\sqrt{3}L_6|p|}{V\lambda_2}A_nA_W + \frac{4\sqrt{3}H_6}{V\lambda_2}A_nA_Y \\ & + \frac{2H_6}{V\lambda_2}A_n^2. \end{aligned}} \tag{4.3}$$

Then the retained R-152 fixed-law chart obeys

$$\mathcal{A}''(0)[H,H] > \left(\frac{7}{50} - C_{2,p}\right)\|H\|_{\mathrm{F}}^2. \tag{4.4}$$

In particular, $C_{2,p} \leq 1/25$ preserves curvature above $1/10$.

Proof. Spatial Cauchy–Schwarz applied to the three R-152 current-mean terms gives

$$\begin{aligned} |T_1| &\le 2\sqrt{3}L_6 be A_n A_W \|H\|_{\mathrm{F}}^2, \\ |T_2| &\le 2\sqrt{3}H_6 b^2 A_n A_Y \|H\|_{\mathrm{F}}^2, \\ |T_3| &\le H_6 b^2 A_n^2 \|H\|_{\mathrm{F}}^2. \end{aligned} \tag{4.5}$$

Here $b^2 \le 2/(V\lambda_2)$ and $e = 2|p|b$. Substitution gives (4.3), and R-152 supplies the 7/50 base gap. □

For normalized RMS quantities $a_n = A_n/\sqrt{V}$, $a_W = A_W/\sqrt{V}$, and $a_Y = A_Y/\sqrt{V}$, the explicit $1/V$ disappears. Also $A_n \le \sqrt{V}N$, $A_W \le \sqrt{V}M_W$, and $A_Y \le \sqrt{V}M_Y$, so (4.3) is no larger than the R-152 essential-supremum collar. Its coefficients are exact for the declared R-130 plus Cauchy–Schwarz envelope; optimality is not claimed.

For a fixed predictable direction H, a weighted condition

$$\mathbb{E}[C_{2,p}\|H\|_{\mathrm{F}}^2] \le \frac{1}{25}\mathbb{E}\|H\|_{\mathrm{F}}^2 \tag{4.6}$$

is sufficient. If (4.6) is demanded for every event-localized predictable H, however, it is equivalent to $C_{2,p} \le 1/25$ almost surely. Therefore the new collar is spatially weaker but not probabilistically weaker as a uniform all-directions theorem.

5. Exact A1 control-mean gradient constant

For the canonical control-generated mean $m_h = Lh_-$, Parseval and $LL^* \preceq A^{-1}$ give

$$\|\nabla m_h\|_2^2 \le \sup_{x\ge 0} \frac{x}{f(x)}\|h_-\|_{\mathcal{S}}^2, \qquad f(x) = x^2 - ax + c, \tag{5.1}$$

where

$$a = \frac{4626377063}{5000000000}, \qquad c = \frac{5020336473}{10000000000}. \tag{5.2}$$

Since

$$\frac{d}{dx}\frac{x}{f(x)} = \frac{c - x^2}{f(x)^2}, \tag{5.3}$$

the exact maximum is

$$c_{\mathrm{grad}} = \frac{1}{2\sqrt{c} - a} = 2.033300406271083\dots. \tag{5.4}$$

The rational certificate

$$\frac{41}{20}f(x) - x = \frac{410000000000x^2 - 579362919166x + 205833795393}{200000000000} \tag{5.5}$$

has discriminant

$$-\frac{476508084992737466111}{10^{22}} < 0. \tag{5.6}$$

Thus

$$\boxed{\|\nabla m_h\|_2^2 \le c_{\mathrm{grad}}\|h_-\|_{\mathcal{S}}^2 < \frac{41}{20}\|h_-\|_{\mathcal{S}}^2.} \tag{5.7}$$

This bound concerns the mean generated by the control and its raw source norm. It does not apply to a revealed Gaussian mean, does not include the 9/20 action coefficient, and does not by itself pay a product with the current direction.

6. Gaussian-past and ultraviolet obstruction

Proposition 6.1 (no deterministic absolute collar for a nondegenerate derivative-active Gaussian past)

Assume one uncancelled revealed root has coefficient $g \sim \mathcal{N}(0, \sigma^2)$ with $\sigma > 0$ and spatial profile φ such that $\|\nabla\varphi\|_\infty > 0$. Condition on all other roots and write

$$m = m_0 + g\varphi, \qquad n = \nabla m. \tag{6.1}$$

Then

$$N(g) \geq |g|\|\nabla\varphi\|_\infty - \|\nabla m_0\|_\infty. \tag{6.2}$$

The R-152 collar contains the strictly positive term $2H_6 N(g)^2/\lambda_2$. Hence

$$\operatorname*{ess\,sup}_g C_{\text{past}}(g) = \infty, \qquad \Pr\{C_{\text{past}} > K \mid m_0\} > 0 \tag{6.3}$$

for every finite K. The same conclusion applies to $C_{2,p}$ when the L^2 derivative profile is nonzero. Therefore the deterministic almost-sure threshold $C \leq 1/25$ cannot close the general Gaussian strict past.

The conclusion is unbounded support and positive-probability violation, not almost-sure violation. Small $|g|$ can satisfy the collar with positive probability. A zero mode, a derivative-inactive root, or an exact signed feedback cancellation lies outside this fixture.

Event localization gives the companion logical boundary. If

$$\mathbb{E}[C_{2,p}\|H\|^2] \leq \kappa \mathbb{E}\|H\|^2 \tag{6.4}$$

holds for every bounded predictable direction and the core admits $H = H_0 \mathbf{1}_E$, take $E = \{C_{2,p} > \kappa + \epsilon\}$. Equation (6.4) fails unless this event has zero probability. Thus an averaged multiplier cannot evade (6.3) for the full predictable class.

There is a separate ultraviolet warning. In the A1 sharp-cube covariance,

$$\operatorname{Tr}\Gamma(k) = \frac{6}{Y_{\text{kin}}|k|^4} + O(|k|^{-6}),$$

so the separated future-current envelope obeys

$$A_{Y,\Lambda}^2 = \sum_{k \in \text{cube}} |k|^2 \operatorname{Tr}\Gamma(k) = \Theta(V\Lambda), \qquad A_{Y,\Lambda} = \Theta((V\Lambda)^{1/2}). \tag{6.5}$$

Indeed the exact max-norm shell has $24r^2 + 2$ lattice points, giving

$$8N \leq \sum_{0 < \|n\|_\infty \leq N} \frac{1}{|n|^2} \leq 26N. \tag{6.6}$$

Indexing and cutoff conventions change the leading constant, not the linear growth for a positive-density future tail. This is the already registered separated-current obstruction, not a divergence of the covariance-normal endpoint: the signed trace cancellation in Theorem 2.1 removes the primitive future-current divergence.

Finally, $B \succeq 0$ does not sign (3.3). In the scalar quadratic row $B(w) = \kappa w^2$, the matrix in the (z, u) variables is

$$\kappa \begin{pmatrix} n^2 - \sigma_<^2 & 2mn \\ 2mn & m^2 + \sigma_X^2 \end{pmatrix}. \tag{6.7}$$

At $m = n = \sigma_X^2 = 1$ and $\sigma_<^2 = 1/2$, its determinant is $-3\kappa^2$. This is a logical sign fixture, not a production counterexample. It explains why the covariance deficit and DB cross terms must remain in one signed operator.

7. Exact remaining route

The final-fresh-root fiber is now explicit. The unresolved T-050 object is larger. A legal continuation must derive, from one scalar action and one predictable chart, all of the following once:

1. the earlier-root future-feedback and covariance-normal future-variance-minus-trace/forest completion;
2. the legal two-sided reverse of the same forward block;
3. balanced/root compatibility and every cross-root sextic block;
4. the genuine complete-low and returned-mean Feshbach block;
5. chart-connection terms whenever the production control map is curved;
6. the cutoff-, refinement-, and control-uniform form bound $K \succeq -4\mathrm{I}/5$, or the finite owner-complete matrix bound $M \succeq \mu G$.

The present calculation forbids three shortcuts: treating the future trace as an additional reserve after (2.8), replacing signed completion by separated absolute current norms, and inferring global positivity from the positive B term alone. It neither proves nor disproves the remaining production Loewner inequality.

8. Devil's-advocate review

1. **Objection: zero field-current covariance is only uncorrelated, not independent. DISMISSED under the declared hypothesis.** The fresh pair is jointly Gaussian, so zero cross covariance gives independence. The result does not extend this inference to non-Gaussian laws.
2. **Objection: m, n, z, u were varied independently. VALID-with-mitigation.** Equations (2.2)–(2.3) retain the production incidences $n = \nabla m$ and $u = \nabla z$. Independent scalar fixtures audit coefficients or logic only and are not production solutions.
3. **Objection: the positive sixth-power Hessian makes the full operator positive. UPHELD against that inference.** It may vanish, and no registered theorem compares it uniformly with the signed covariance deficit and cross terms.
4. **Objection: the spatial L^2 collar solves the random-past problem by averaging. UPHELD.** It improves only the spatial norm. For all predictable directions, event localization restores the almost-sure multiplier condition.
5. **Objection: Gaussian support proves the collar is violated almost surely. UPHELD against that stronger wording.** The proved statement is unbounded essential support and positive-probability violation.
6. **Objection: the $41/20$ bound pays the full source budget. UPHELD.** It bounds only the control-generated mean gradient in the raw source norm. It is not the $9/20$ action payment and does not control the revealed past.
7. **Objection: $A_Y \sim \sqrt{\Lambda}$ proves the PDE or A7 composite diverges. UPHELD against that inference.** Only the separated absolute current envelope diverges. The signed future-current/trace cancellation is exact.

8. **Objection: positive curvature proves a local minimum. UPHELD.** The first variation (3.2) and chart connections may be nonzero.

9. **Objection: this result selects BCC, a uniform phase, or another PDE. UPHELD against every such inference.** R-153 is phase-neutral conditional mathematics and supplies no physical model-selection verdict.

External review is invited for the conditional independence step, the trace sign in (2.8), every $1/2$ and factor-two coefficient in (3.2)–(3.3), the sixth-power factors, the $1/V$ normalization in (4.3), the discriminant in (5.6), and the quantifiers in Proposition 6.1.

9. Reproduction and Result footer

The primary route differentiates a polynomial endpoint with exact symbolic arithmetic and derives every A1/R-130 constant from pinned upstream data. The independent route uses only the Python standard library, rebuilds the endpoint as a two-variable rational polynomial, and independently checks the lattice, gradient, collar, connection, and sign certificates. The integrated verifier reruns both children, fails closed on authority and artifact hashes, rebuilds the PDF deterministically, scans it for active content, extracts required scope text, renders every page with Poppler, and checks all registered surfaces.

```
Result ID: A13-CLASSII-PRODUCTION-STRICT-PAST-CONDITIONAL-HESSIAN-
  WEIGHTED-COLLAR-BOUNDARY (R-153).
Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION.
Precise statement: On the declared final fresh production root, (2.5) and
  (3.2)-(3.8) give the exact strict-past endpoint, variations, and owners;
  (4.3) and (5.4) give conditional collar and control-mean bounds; Gaussian
  support forbids a uniform deterministic absolute collar. T-050 remains open.
Scope: Fixed cutoff/floor, retained antipodal p:2p chart, strict-past fixed
  common-even Gaussian law, zero same-point field-current cross covariance,
  actual A1 incidences, and the covariance-matched A7 trace.
Dependencies: A1, A7, R-130, R-150--R-152, and Sections 1-2 hypotheses.
Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/a13_classii_production_strict_past_conditional_hessian_
  weighted_collar_boundary_verify.py
Expected output: Integrated 177/177 PASS, including primary 33/33,
  independent 27/27, 60 embedded child rows, 117 integrator-only checks, and
  every artifact/PDF/generated-surface gate.
Falsification gate: Failure of any cancellation, derivative factor, collar or
  gradient coefficient, Gaussian-support quantifier, authority hash,
  executable/JSON/PDF artifact, exploration, negative, or public-surface pin.
Tier before / after: T4 / T4; no promotion.
No-overclaim statement: No complete progressive Hessian, production Loewner
  gap, T-050/A13 closure, Nelson/interacting measure, phase selection, PDE
  validation/replacement, or Sector-A closure follows.
Next required action: Reopen T-050 only from one actual predictable production
  chart deriving every feedback/forest/reverse/balanced/low/sextic/connection
  owner once; otherwise continue phase-neutral T-054 model selection.
```